A DESIGN UNBIASED VARIANCE ESTIMATOR OF THE SYSTEMATIC SAMPLE MEAN

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DECLARATION

This research proposal is my own work and has not been presented elsewhere for a degree award.

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Declaration by supervisors.

This research proposal has been submitted for examination with our approval as university supervisors.

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DEDICATION

To my Mum,Dad,Sisters and Brothers.

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ABSTRACT

Systematic sampling is normally used in surveys of finite populations due to its appealing simplicity and efficiency. When properly applied, it can reflect stratification in the population and thus can be more precise than simple random sampling. In systematic sampling, the sampling units are evenly spread over the entire population. However, this sampling scheme is sensitive to correlation between units in the population. A positive autocorrelation decreases the precision while a negative autocorrelation will improve the precision compared to simple random sampling. The problem of this technique is that it is not possible to estimate the design variance without bias. This study proposes an estimator for the design variance based on a non-parametric model for the population using local polynomial regression as the estimation technique. The non-parametric model is more flexible that it can hold for many practical situations.A simulation study was carried out to enable the comparison of the efficiency of the proposed estimator to the existing ones.The performance measures used include: Relative Bias(RB) and Mean Square Error(MSE). The results showed that the local polynomial estimator based on nonparametric model is consistent and design unbiased for the variance of systematic sample mean.The simulation study gave smaller values for the relative biases and mean squared errors for proposed estimator.

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Chapter 1

INTRODUCTION

1.1 Background of study

Systematic sampling is a probability sampling technique where a sample is obtained by selecting everyk^{th} element of the population where k is an integer greater than 1. The first number of the sample must be selected randomly from within the first k elements. The selection is done from an ordered list. It is a popular method of selection especially when units are many and are serially arranged from 1 to N. Suppose that N the total number of units is a multiple of the required sample size n and an integer k, such that $N = nk$, a random number is selected between 1 and k. A sample which comprises of the first unit is selected randomly and every K^{th} unit, until the required sample size is obtained. The interval k divides the population into groups. In this method we are selecting one cluster of units with probability$\frac{1}{K}$. Since the first number is drawn at random from 1 to k, each unit in the supposedly equal clusters gets the same probability of selection $\frac{1}{K}$.

Systematic sampling is widely used in surveys of finite populations such as forest where other sampling scheme cannot be easily applied. This is due to its appealing simplicity and efficiency. When properly applied, the method picks up any obvious

or hidden stratification in the population and thus can be more precise than simple random sampling. Also, systematic sampling is easy to implement, thus reducing costs.

Since a systematic sample can be regarded as a random selection of one cluster, it is not possible to give an unbiased or even consistent design based estimator of the variance and this is the challenge faced by researchers who apply it in practice. There are two approaches that are proposed to solving this problem. One is to postulate a super-population model characterizing the population structure and to obtain model-unbiased estimators of variance Cochran [1977]. The super-population model is used to describe the relationship between the auxiliary variable and the study variable. This approach may not yield satisfactory results because the model assumption is usually hard to verify in practice and the unbiased estimators of variance can be sensitive to model assumption. The second approach is to take additional observations (supplementary sample) typically of smaller size than the first sample via simple random sampling Zinger [1980] or systematic sampling.

Nonparametric regression is motivated by the fact that it provides a flexible way of studying the relationships between variables and also results in good estimators thus increasing their efficiency compared to estimators obtained using designed based approaches.

In this framework,the study is concerned with the estimation of the variance of systematic sample mean using a nonparametric approach with the aid of a super-population model. It also offers the methodology to study the convergence properties of the proposed estimator.

1.2 Problem statement

Variance estimation for systematic sample mean still remains an issue that has not been addressed as only estimation procedures which are not so robust have been proposed. In view of this, exact computation of a robust estimator for variance in the systematic sample mean or total mean is an open area of research.

1.3 Justification of the study

Systematic sampling is one of the survey techniques that have been applied in many fields, including both artificial and natural population. This is normally in situations where other sampling techniques cannot be applied. A well-known and long-standing issue in surveys that follow a systematic sampling design is the lack of a design-based variance estimator. Because of this, there have been attempts to adjust the sampling design itself to allow for variance estimation, including for example the drawing of multiple systematic samples and the drawing of a partial systematic sample supplemented with a simple random sample Zinger [1980]. This type of solutions are only rarely used, however, and the majority of applications continue to apply 'pure' systematic sampling combined with a not fully satisfactory variance estimator. Wolter [2007] discuss various estimators for the systematic sample mean but they are all biased for the design variance.

Much of modern sampling practices rest on the processes of selecting individuals at random or according to certain systematic procedures. Systematic sampling which employs a simple rule of counting cases in some convenient order and selecting for the sample every k^{th} case or of using some similar pattern at measured intervals. This technique has been employed in physical and engineering sciences, other applications have been made in forestry, agriculture, animal husbandry and meteorology. In these entire surveys one will want to estimate unbiased estimate for variance to assist in

the construction of confidence intervals and also estimation of mean square errors to help make accurate inferences about the population under study.

1.4 OBJECTIVES OF THE STUDY

1.4.1 General objective

To use Regression Techniques in designing an unbiased variance estimator of the systematic sample means.

1.4.2 Specific objectives

1. To use Local Polynomial Regression to propose an estimator for variance of the systematic means.
2. To compute the Mean Square Error of the proposed estimator.
3. To compute the Relative Bias of the proposed estimator.
4. To compare the efficiency of proposed estimator to $\widehat{V}_{SRS}, \widehat{V}_{OL}$ and $\widehat{V}_{NO}$

1.5 OUTLINE OF THE PROJECT

The organization of this study is as follows: Chapter 2 discusses the literature review on the estimators of variance of systematic sample mean,trade-off between bias and variance. In chapter three, the methodology used in building the proposed estimator is discussed. Here,the proposed estimator is given and the proof of convergence property of this estimator is also discussed here. Finally in this chapter, the procedure on how simulation study will be carried out is also explained.
In chapter four, the researcher discusses the simulation results and in the last chapter, the conclusions and recommendations from this study is given.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

Systematic sampling is widely used in surveys of finite populations such as forest where other sampling scheme cannot be easily applied. This is due to the fact that it is easily to implement, and hence turns out to be more efficient with respect to other sampling schemes when there is some hidden stratification in the list of population units Cochran [1977]. However, under this sampling scheme, there is no design-unbiased estimator of the variance of the sample mean (Iachan, 1982).

Several approaches have been proposed to solve this problem including taking supplementary samples typically of smaller size than the first sample via SRS Zinger [1980] and develop an unbiased estimator for the variance. The second approach is to postulate a superpolation model and obtain a model unbiased estimator of variance Cochran [1977]. For the later approach, one can assume a linear model or a quadratic model for the data, but the estimator based on the linear model can fail if the model is misspecified Montanari.G and Bartolucci.F [1998]. Non parametric regression provides a solution to the challenge experienced when the model is misspecified as it provides a flexible way of studying the relationship between variables

and also results in good estimators thus increasing their efficiency.This study uses non parametric regression technique that is local polynomial regression to develop a design unbiased and consistent estimator for the variance of systematic sample mean.

2.2 Local polynomial regression estimators

Nonparametric methods are rich classes of statistical tools that have gained acceptance in most areas of statistics. They can be used analyze data,estimate trends and conduct inferences without having to fully specify a parametric model for the data (see Breidt and Opsomer 2009).

Non parametric regression methods have been recently applied to the estimation of the regression functions in many different area of research and particularly in survey research. In particular, local polynomial regression kernel smooth Wand and Jones [1995] is well known due to the good theoretical and practical properties. In other words, the nonparametric techniques has become an attractive method in survey sampling since this methods only works under the assumption that the regression function in superpopulation model is smooth. Local polynomial regression has gained popularity in literature in the estimation procedure of the survey parameters such as population totals, means and variance.In systematic sampling where there is lack of a designed unbiased estimator for systematic sample mean, local polynomial regression estimator based on model-based approach have been developed which have been found to be consistent and unbiased.

Breidt and Opsomer [2000] **?** consider nonparametric models for a working model $\mathcal{L}$ in a model-assisted framework to improve the design properties of estimators, and they introduce nonparametric model-assisted estimators for the finite population total based on local polynomial regression. The actual efficiency of model-assisted estimators, that is, their design variance, depends on the goodness of fit of the model

$\mathcal{L}$,even if the design properties of model-assisted estimators are not dependent on the model assumptions.

For a linear superpopulation model, Montanari.G and Bartolucci.F [1998] proposed a model based variance estimator using OLS which was approximately unbiased for the variance of systematic sample mean under linear super-population model. However this estimator lacked some accuracy and efficiency due to a higher contribution of the bias if the systematic component of the super-population is significantly different from linear. Later Montanari.G and Bartolucci.F [2006] proposed a new class of unbiased estimators of the variance of systematic sample mean which included some simple nonparametric estimators under the assumption that the population follows a superpopulation model that satisfied some mild assumptions. They showed that, the estimator based on local polynomial regression as the estimation technique under the assumption that the population follow a linear trend and the errors are homoscedastic and uncorrelated.The simulation results showed that the LPR estmator performed better in terms of relative bias and mean square error as they were all small.

X.Li [2006] using the work that was done by Montanari.G and Bartolucci.F [2006] proposed a model based variance estimator using local polynomial regression as the estimation approach based non parametric superpopulation model with homoscedatic errors. The simulation results suggest that the nonparametric approach using local polynomial regression was an effective method of estimating the variance of systematic sample mean under a range of population model specifications. In his work he demostrate that, the proposed estimator was both asymptotically unbiased and design consistent under mild set of assumptions. The results from the simulation experiment showed that the proposed nonparametric estimator based on local polynomial regression performed well relative to other estimators interms of relative

bias and mean square errors.

X.Li and Opsomer [2010] and Ayora.O [2014] also using the work that was proposed by Montanari.G and Bartolucci.F [2006] and X.Li [2006] considered a broadly applicable model for the data, in which both the mean and the variance are left unspecified subject only to smoothness assumptions.they then proposed a model-based nonparametric variance estimator, in which both the mean and the variance functions of the data are estimated nonparametrically using local polynomial regression as the smoothing technique. The simulation experiment performed showed that the this estimator perform better giving small relative bias and mean square errors as compared to the other classical estimators discussed in Wolter [2007].

This study considers a more applicable model in which the mean function is unspecified but the variance function is homoscedastic. The study proposes a model-based nonparametric estimator using local polynomial regression as the smoothing technique for variance of systematic sample mean. The study later shows that the proposed estimator is model consistent for the design variance of the survey estimator, subject only to the population smoothness assumptions.

2.3 Trade-off beween Bias and variance

The choice of the bandwidth, h is of crucial importance tool for local polynomial regression. Smaller bandwidth results in less smoothing while larger bandwidth oversmooths the curve. There is a trade-off between variance and bias. Large values of bandwidth will reduce the variance since more points will be included in the estimate. However, as the the bandwidth increases, the average distance between the local points and x_0 will increase. This can result in a larger bias in the estimator.
A natural way to choose a bandwidth and balance this trade-off is by minimizing

the mean square error(MSE) Fan and Gijbels [1996]. Therefore one should choose an optimal bandwidth to minimize MSE so as to balance the trade-off between the bias and variance.

In addition to choosing the optimal bandwidth, it is also important to choose the appropriate order of polynomial to fit as when choosing a bandwidth, there is also trade-off between bias and variance. Higher order polynomials allows for precise fitting meaning the bias will be small but the order increases, so does the variance. However this increase is not constant. The asymptotic variance for $\hat{\mu}(x)$ only increases whenever the order goes from odd to even. There is no loss when going from $p = 0$ to $p = 1$ but going form $p = 1$ to $p = 2$ will increase asymptotic variance. This suggests only considering odd-ordered polynomials since the gain in bias appear to be free with no associated cost in variance Fan and Gijbels [1996],Wand and Jones [1995]

Fan and Gijbels [1996] suggests an addaptive method of choosing the correct order of polynomial based on local factor, allowing p to vary for different points in the support of data. The resulting estimator has the property of being robust to bandwidth. This means that if the chosen bandwidth is large is too large,a higher order polynomial is chosen to better model the boundaries of the data. If the chosen bandwidth is too small, a lower order polynomial is chosen to help make the estimate numerically stable and reduce the variance. Therefore one should select an appropriate bandwidth and order of the polynomial to balance the trade-off between the bias and variance in order to give an appropriate amount of smoothing.

2.4 Relative Bias and Mean Square Error

In many applications, the survey statistician will wish to emphasize efficient estimation of $\overline{Y}$ and thus will prefer single-start systematic sampling to the supplementary

techniques. The statistician will also wish to employ a fairly robust variance estimator with good statistical properties, for example small bias and MSE and good confidence interval coverage rates.

The Bias of an estimator $\hat{\theta}$ of a parameter θ if the difference between the expected value of $\hat{\theta}$ and θ that is

$$Bias(\hat{\theta}) = E(\hat{\theta}) - \theta \tag{2.1}$$

If the Bias is identically equal to zero the estimator is said to be unbiased that is $E(\hat{\theta}) = \theta$for all θ. The Bias of an estimator measures the accuracy of an estimator. The relative Bias is the computed as

$$RB(\hat{\theta}) = \frac{Bias(\hat{\theta})}{\theta} \tag{2.2}$$

and it provides a measure of the magnitude of the bias. The larger the relative bias, the more the biased an estimator is. Smaller values of relative bias are therefore desired.

On the other hand Mean Square Error can be defined as

$$MSE(\hat{\theta}) = E_\theta(\hat{\theta} - \theta)^2 = var(\hat{\theta}) - (Bias(\hat{\theta}))^2 \tag{2.3}$$

.

Therefore, the mean square error has two components, one measures the variability of the estimator(precision) and the other measures the bias(accuracy). An estimator that has good MSE properties has small combined variance and bias. Thus, Estimators with small values of MSE are generally preferred to those with largest values of MSE.

Given two estimators $\hat{\theta}$ and $\hat{Q}$ of a parameter θ, then, the relative efficiency of $\hat{\theta}$

relative to $\hat{Q}$ is defined by

$$RE(\hat{\theta}, \hat{Q}) = \frac{MSE(\hat{\theta})}{MSE(\hat{Q})} \tag{2.4}$$

Cochran [1977] compares different estimators for variance of systematic sample mean using the Relative Bias and MSE. In his study through simulation experiment, the relative bias and MSE of different estimators are computers and those estimators with smaller values of RB and MSE are considered to be better. Using the same technique of RB and MSE,Montanari.G and Bartolucci.F [2006] simulation results shows that the linear estimator for variance of systematic sample mean had smaller values of mean square error and RB compared to those estimator discussed in Wolter [2007]. X.Li [2006] proposes another estimator based on Nonparametric regression and this estimator had smaller Relative bias and MSE under different model specification.The simulation result showed that this estimator can perform well under different Superpopulation model if the correct bandwith is provided.

Local polynomial estimators have high minimax efficiency among the class of linear smoothers, including those produced by kernel,orthogonal series, and spline methods, both in the interior and at the boundary. Fan [1992],Fan [1993] discusses in detail the local linear fit in comparison with the local constant fit and shows that the local linear regression smoothers have the desirable mean squared error (MSE), the design adaptation property,no boundary effects, and high asymptotic minimax efficiency properties.Fan [1992] prove that the local linear regression estimator adapts automatically to estimation at the boundary and give expressions for the conditional MSE and mean integrated squared error (MISE) of the estimator.

Chapter 3

METHODOLOGY

3.1 Introduction

This chapter gives the methodology used in the developing the proposed estimator. Section 3.2 we introduce appropriate notations and then decsribe the systematic sampling. Section 3.3 reviews the properties of local polynomial regression and section 3.4 reviews the variance estimation techniques that are used currently. Section 3.6 adopt the assumptions that are used to develop our proposed estimator. Section 3.7 describes the proposed estimator and its properties. Section 3.8 describes the simulation experiment that is carried out to compare the proposed estimator with other existing estimators.

This study is concerned with the estimation of the variance of systematic sample mean using a nonparametric approach(local polynomial regression) with the aid of a superpopulation model.It also offers the methodology to study the convergence properties of the proposed estimator.

In the current study,let $Y_j \epsilon R(j = 1, 2, \ldots)$ be finite population measurements of size N representing some survey characteristics and $X_j \epsilon R(j = 1, 2, \ldots)$ be a vector of auxiliary variables which is considered fixed and $j = 1, 2, \ldots, N$. Let k be the sampling interval and $p(.) = \frac{1}{k}$ be the probability of each element in the sample

being selected from the population,then the systematic sample will consist of the observations $i, i+k, \ldots, i+(n-1)k$ where n is the sample size and the ith systematic sample will be $y_i, , \ldots, y_{(n-1)(k+i)}$.

Let $\overline{Y}$ be the population mean and $\overline{y}_i$ be the ith systematic sample mean. The parameter to estimate is the variance of $\overline{y}_i$ which is defined by equation 3.3

To estimate this variance, the local regression smoothing function defined by equation 3.8 estimated from the ith sample is used where $W_{ij} = diag(K(\frac{x_i - x_j}{h})), i = 1, 2, \ldots, n$ with h being the smoothing parameter and $K(.)$ a kernel function.

3.2 Review of Systematic Sampling

Suppose that the population size is N units and that the study variable Y_i where$i = 1, 2, 3, 4, \ldots, N$. Then the population mean is given as

$$\overline{Y} = \frac{1}{N} \sum Y_i \tag{3.1}$$

To draw a systematic SYS we first sort the population using some criterion. For example we can sort by one of the auxiliary variables in X_i. If the study variable Y and auxiliary variable X are related through a certain function, sorting by X may provide a good spread of $Y's$ so that a systematic sample can pick up hidden structures in the population. If we sort the population by some criterion that is not related to Y at all, for instance sort by a variable Z which is independent of Y, then we will have a random permutation of the population. In this case systematic sampling is equivalent to $SRSWOR$. After sorting the population we randomly choose an element from the first k ones say the ith one, then this systematic sample consists of the observations $i, i+k, \ldots, i+(n-1)k$. Thus Systematic sampling amounts to the selection of a single complex sampling unit that constitutes the whole sample. A systematic sample is a srs of one cluster unit from a population of k cluster units. The table 3.1 illustrates this procedure. Each column corresponds to a possible

sample number							
1	2	3	...	...	i	...	k
y_1	y_2	y_3	...	...	y_i	...	y_k
...	...	...		...	...	...	...
...	...	...	...	...	...	...	...
$y_{(n-1)(k+1)}$	$y_{(n-1)(k+2)}$	$y_{(n-1)(k+3)}$	...	...	$y_{(n-1)(k+i)}$	...	y_{nk}
$\overline{y}_1$	$\overline{y}_2$	$\overline{y}_3$	...	...	$\overline{y}_i$	...	$\overline{y}_k$

Table 3.1: Composition of the k Systematic Samples

sample systematic sample. The interval k divides the population into n rows of k elements each. One element from each row is selected and each element has the same location on each row.

Each element in the population has a probability $\frac{1}{k}$ of being selected from the population. The population mean is estimated by the ith sample mean

$$\overline{Y}_s = \frac{1}{n}\sum_{i=1}^{n} Y_i \tag{3.2}$$

The design based variance for this mean was first derived by Madow and Madow [1944] and is give by

$$Var_p\left(\widehat{Y}_s\right) = \frac{1}{k}\sum_{i=1}^{k}\left(\overline{Y}_s - \overline{Y}\right) \tag{3.3}$$

But there is no unbiased design based estimate of $Var_p(\overline{Y}_s)$ for the general variable Y. Among the eight estimators evaluated by Wolter [2007] as the estimates of$Var_p(\overline{Y}_s)$ we look at the three main ones which are used in practice. One of the approaches is to treat the systematic sample as if it had been obtained by SRS. This estimator is defined as

$$\widehat{V}_{SRS} \quad = \tfrac{1-f}{n}\tfrac{1}{n-1}\textstyle\sum_{i\epsilon s}(Y_i - \overline{Y}_s)^2 \tag{3.4}$$

where

$$f = \frac{n}{N}$$

The other two estimators are based on pairwise differences and are recommended in Wolter [2007] as the best general purpose estimators of $Var_p(\overline{Y}_s)$, these estimators are defined as

$$\widehat{V}_{OL} \quad = \tfrac{1-f}{n}\tfrac{1}{2(n-1)}\textstyle\sum_{i=2}^{n}(Y_i - Y_{i-1})^2 \tag{3.5}$$

Which uses all successive pairwise differences and hence uses OL The other estimator is defined by

$$\widehat{V}_{NO} \quad = \tfrac{1-f}{n}\tfrac{1}{n}\textstyle\sum_{i=1}^{\frac{n}{2}}(Y_{2i} - Y_{2i-1})^2 \tag{3.6}$$

This takes on successive NO The three estimators are designed biased for $Var_p(Y_s)$ in general. The first estimator $\widehat{V}_{SRS}$ is viewed suitable when the ordering of the population is thought to have no effect on $\overline{Y}_s$ or is considered as a conservative estimator when the ordering is related to the variable Y. However, as discussed by X.Li and Opsomer [2010], the unbiasedness of $\widehat{V}_{SRS}$ for uninformative ordering only holds if one averages over samples and over orderings of the population, so not design strictly design unbiasedness. The bias of $\widehat{V}_{SRS}$ for a fixed ordering of the population can be larger and either positive or negative. The last two estimators tended to have smaller bias in the simulation experiments reported in Wolter [2007]. To obtain an unbiased estimate of $var_p(\overline{Y}_s)$, the following three designs have to be considered.

1. Multiple systematic sampling using a randomly determined starting position for each systematic sampling stage.

2. Systematic stratified - Two or more systematic samples (each with a different random start position) are taken within each stratum

3. Two stage sampling where the sub samples are collected according to systematic sampling design

4. Complementary systematic and random sampling where a systematic sample is supplemented by a random sample of size m from the remaining population units.

3.3 Review of properties of local polynomial regression

Nonparametric regression has become a rapidly developing and growing field of statistics. Nonparametric approaches to regression are flexible and data-analytic ways to estimate the regression function without the specification of a parametric model, that is, to let data find a suitable function that well explains the data. Some of the commonly used methods to obtain nonparametric regression estimators are based on kernel, spline and orthogonal series functions. Local modeling techniques with kernel weights provide a basic and easily understood nonparametric approach to regression. Local polynomial regression is a generalization of kernel regression since the regression function at a point x in kernel regression is estimated by a locally weighted average, which can be shown to correspond to fitting degree zero polynomials, that is, local constants. Wand and Jones [1995] give a clear explanation of kernel smoothing including local polynomial regression.

Let K denote the Kernel function and h the bandwidth. Consider a single univariate auxilliary variable x, then the nonparametric model is given by

$$y_i = \mu(x_i) + \varepsilon_i \; i = 1, \ldots, N \tag{3.7}$$

where

$$E(\varepsilon_i) = 0$$

$$Var(\varepsilon_i) = v(x)$$

$\mu(x_i)$ is the smooth mean function of x and $v(x)$is a smooth and strictly positive. let $K_h(x) = \frac{1}{h}K(\frac{x}{h})$ and $Y = (y_1, ...y_N)'$. Then, the local polynomial kernel estimator of $\mu(x_i)$ based on the entire population is given by

$$\mu_i = (1, 0, \ldots, 0)(X'WX)^{-1}XWY \tag{3.8}$$

where

$$X = [1, x_j - x_i, ..., (x_j - x_i)^p]_{j\epsilon U}$$

and

$$W = diag(K_h\left(\frac{x_j - xi}{h}\right)$$

μ_i is normally unknown population quantity and therefore it is estiamted from the sample design estimator given by

$$\hat{\mu_i} = (1, 0, ..., o)(X_i'W_iX_i)^{-1}X_iW_iY_i \tag{3.9}$$

where $i\epsilon s$

Local polynomial regression has several advantages of other smoothing techniques. For instance,the method is readily adapted to random and fixed designs, to highly clustered and nearly uniform designs, and to both interior and boundary points. Local polynomial regression estimators do not have boundary bias, that is, they adapt automatically to the boundary, and hence do not require any modifications for correcting the large bias problem at the boundary. Local polynomial estimators

have high minimax efficiency among the class of linear smoothers, including those produced by kernel, orthogonal series, and spline methods, both in the interior and at the boundary. Fan [1992], Fan [1992] discusses in detail the local linear fit in comparison with the local constant fit and shows that the local linear regression smoothers have the desirable mean squared error (MSE), the design adaptation property, no boundary effects, and high asymptotic minimax efficiency properties. Fan [1992] prove that the local linear regression estimator adapts automatically to estimation at the boundary and give expressions for the conditional MSE and mean integrated squared error (MISE) of the estimator. Wand and Jones [1995] extend the results of Fan [1992] on asymptotic bias and variance to the case of local polynomial estimators. Fan and Gijbels [1996] put emphasis on methodologies with a particular focus on applications of local polynomial modelling techniques to various statistical problems including least square regression, quantile and robust regression, survival analysis, generalized linear models and nonlinear time series.Breidt and Opsomer [2000] apply local polynomial regression to model-assisted survey sampling.

One of the important issues in nonparametric regression is the choice of the smoothing parameter, often called the bandwidth. In practice, the bandwidth is often selected subjectively by eye, but there are many situations where it is necessary to have the bandwidth automatically chosen from the data. In data-driven smoothing parameter selection, all methods try to estimate the optimal bandwidth value, which minimizes the mean squared error (MSE) at a point x or the MSE over all values of x, that is, the mean integrated squared error (MISE) of the estimator. Most bandwidth selection methods attempt to find a value for the MISE-minimizing bandwidth, and thus those are called global bandwidth selection methods. Cross-validation (CV) is a well-known method to optimize the smoothing parameter, using the idea of leave-one-out prediction. However, the bandwidth computed by the CV method is very variable and tends to undersmooth in practice, that is, selected bandwidths tend to

be too small. For linear smoothers, calculation of the CV method is easy since the expression of the leave-one-out predictor is a linear function of the complete data predictor. Another approach to bandwidth selection is to estimate MISE directly based on the data. This method estimates the bias and the variance of the estimator and minimizes the estimated MISE with respect to the bandwidth. This "plug-in" method is used most commonly in kernel regression and local polynomial regression. Plug-in approaches give more stable performance. The theoretical choice of a global variable bandwidth based on the plug-in procedure for the local linear smoothers was discussed by Fan [1992]. Wand and Jones [1995] developed a simple direct plug-in bandwidth selector for local linear regression that is seen to work well in practice for a wide variety of functions and is shown to have appealing theoretical and practical properties. Fan and Gijbels [1995] propose a data-driven variable bandwidth selection procedure based on a residual squares criterion and show that local polynomial fitting using the variable bandwidth has spatial adaptation properties.

There is much interest in extending nonparametric methods to the multivariate regression case since parametric modeling is not suitable for fitting regression surfaces well to multivariate data that are so common in many practical applications. The extension of the univariate nonparametric regression techniques to the multivariate setting has some problems. In higher dimensions, a full bandwidth matrix requires a larger number of smoothing parameters and hence bandwidth selection becomes more complicated. The sparseness of data in higher-dimensional space makes nonparametric regression impractical. This problem, known in literature as the curse of dimensionality, means that neighborhoods with a fixed fraction of the points can become very big as the dimensions increase, so that in higher-dimensional settings, nonparametric regression techniques do not perform well for practical sample sizes. Several approaches have been proposed to deal with this problem. The additive modeling methodology as a dimensionality reduction alternative provides a simple

solution. The general regression function is modeled as a sum of univariate functions, one for each prédit or. Nonparametric regression techniques are used to estimate each univariate function. Iterative procedures known as backfitting, which smooth partial residuals, are used to fit the additive model.

Ruppert and Wand (1994) study the asymptotic bias and variance of multivariate local regression estimators. These estimators are shown to have optimal rates of con vergence in Stone [1980] and have proved to be very useful in modeling real data ClevelandW.S and DevlinS [1988]. OpsomerJ.D and Ruppert [1997] provide some theoretical properties of the bivariate additive model when the additive model is fitted by local polynomial regression.

3.4 Review of Variance estimation procedures

In the cases where there is no exact methods that exists for the calculation of the unbiased estimates of the standard errors of the point estimates, the only alternative is to approximate the required quantities. But to achieve these unbiasedness, we impose some simplifying assumption(with respect to the sample design or the statistic to be variance estimated). Another alternative approach is based on the on replication methods. In this study we look at some of the variance estimation procedures under replication methods

3.4.1 The Jackknife estimator

The central idea of jackknife is dividing the sample into disjoint parts, dropping one part and recalculating the estimates of interest based on that incomplete sample. The 'dropped part' is re-entered in the sample and the process is repeated successively until all parts have been removed once from the original sample. These replicated statistics are used in order to calculate the corresponding variance by finding the average of these calculations. Given a sample of size N the jackknife

estimate is found by calculating averaging the estimates of each $N-1$ estimates in the sample. The Jackknife variance estimator of $\overline{Y}$ is then given by

$$\widehat{V_J} \quad = (\tfrac{n-1}{n}) \textstyle\sum_{i\epsilon s} (\overline{Y}_j - \overline{Y})^2 \tag{3.10}$$

$$j = 1, 2, \ldots, n$$

3.4.2 Bootstrap estimator

It involves drawing a series of independent samples from the sampled observations using the same sampling scheme as the one by which the initial sample was drawn from the population and calculating an estimate for each of the bootstrap samples. The Rao - Wu bootstrap may be described as follows

1. let n^1be the bootstrap sample size. Let

$$z_i = \overline{y} + \sqrt{C}(y_i - \overline{y})$$

$$i = 1, 2, 3, \ldots, n$$

where

$$C = \frac{n^1(1-f)}{n-1}$$

2. Draw a simple random sample $\{z_i^*\}_{i=1}^{n^1}$ with replacement from $\{z_i\}_{i=1}^{n}$ let $\overline{z}^* = \sum_{i=1}^{n^1} \frac{z_i^*}{n^1}$ 3. Repeat (ii) a large number of times say B times to get $\overline{z}^*_{(1)}, \ldots, \overline{z}^*_B$ The bootstrap variance estimator is given as

$$\widehat{V}_{RW} \quad = \tfrac{1}{B-1} \textstyle\sum_{b=1}^{B} (\overline{z}^*_{(b)} - \overline{z}^*_{(.)})^2 \tag{3.11}$$

where

$$\overline{z}^*_{(.)} = \sum_{b=1}^{B} \frac{\overline{z}^*_{(b)}}{B}$$

The estimate $\widehat{V}_{RW}$ is a consistent estimator of

$$(1-f)\frac{s_y^2}{n}$$

where

$$s_y^2 = \frac{1}{n-1}\sum_{i\epsilon s}(Y_i - \overline{Y})^2$$

3.4.3 Balanced repeated replication method

These method has a very specific application in cluster designs where each cluster has exactly two final stages units or in case with large number of strata and the only two elements per stratum. It aims at selecting a set of samples from the family of 2^k samples, compute an estimate for each one and then use them for the variance estimation in a way that the selection satisfies the balance property.

3.4.4 Random group method

This procedure consists of drawing a number of samples(sub samples) from the population, estimating the parameter of interest for each sub sample and assessing its variance based on the derivations of these statistics from the corresponding statistics derived from the union of all the sub samples (see Wolter [2007]). The procedure can be distinguished into two main variations based on whether the sub samples are independent or not. In the case of independent random groups, it provides unbiased linear estimates though small biases may occur in the estimation of non linear statistics. In the case of dependent random groups, a bias is introduced in the results which however tends to be negligible for large - scale surveys with small sampling fraction.

3.5 Review of the Order algebra

The Landau's symbols (Big O and Little o) comes from the name of the German number theoretical Edmond Landau who invented the Big O and the Little o notations. The latter O and o is used because the rate of growth of a function is also called its order. Suppose a_n be a sequence of real numbers. then $a_n = O(1) \Leftrightarrow a_n$ is bounded. The stochastic boundedness means that $\exists c > 0$with $|a_n| < c$ for all n. More generally,let a_n and b_n be a sequence of real numbers, then

$a_n = O(b_n) \Leftrightarrow a_n = r_n b_n$ with $r_n = O(1)$ let $f : R \to R$,then

$f(h) = O(1)$ as $h \to 0$ which means that, for any sequence $h_n \to 0$ as $n \to \infty$, the sequence $f(h_n)$ is bounded. This means that, there exist a constant c>0 with $|f(x)| < c$ for all $x\epsilon[-\epsilon, \epsilon]$ for some $\varepsilon > 0$.

Applying this concept of oder algebra to probability, a sequence of random variables X_n is said to converge in probability at a rate R_n given as $X_n = O_p(R_n)$, which means that, $X_n = Y_n R_n$ with Y_n converging to 0 in probability. Therefore

$O(1)$ is a constant. This means that the sequence converges to zero

$O(n)$ linear. The sequence converges or the rate of growth is in a linear form

$O(n^2)$ quadratic. The rate of convergence take a quadratic form.

$O(n^c)$ polynomial. The rate of convergence take the form of a polynomial.

for the case of little o

$f(x) = o(g(x))$ means that f grows much slower than g and is insignificant in comparison

This is Equivalent to

$lim_{x\to\infty} \frac{f(x)}{g(x)} = 0$

3.5.1 The Linear model and use of order algebra in them

In the model based approach under assumption A_1in section 2.3 above the variance of systematic sample mean can be estimated as follows. Let $Y_j \epsilon R(j = 1, 2, \ldots)$ be

a set of independent and identically distributed random variables. Let $X_j \epsilon R^d (j = 1, 2, 3, \ldots)$ be vectors of auxiliary variables, which we consider as a fixed. Suppose the linear super-population model, denoted by L, is given as

$$Y = X\beta + e \tag{3.12}$$

where $E_L(e) = 0$ and $Var_L = \sigma^2\Omega$ where $E_L$and Var_L denote the expectation and variance under model L. We assume the errors to be mutually independent that is Ω is a diagonal matrix and $\Omega = diag(w_1.....w_N)$ each element of the Ω is assumed to be known. In model 3.12 above

$$Y = (y_1, y_2, \ldots, y_N)^T$$

$$\beta = (\beta_0, \beta_1, \ldots, \beta_d)^T$$

we can now write the design variance 3.3 as

$$Var_p(\overline{Y}_s) \quad = \tfrac{1}{kn^2} Y^T N Y \tag{3.13}$$

where $N = M^T H M$ with $H = I_n^T \bigotimes 1_k$ here $\bigotimes$ is the Kronecker product and $1_k is$ is a column vector of $1's$ of length k Then the model anticipated variance is

$$E_L(Var_p(Y_s)) \quad = \tfrac{1}{kn^2}\beta^T N \beta + \tfrac{1}{kn^2} tr(N\Omega)\sigma_L^2 \tag{3.14}$$

Montanari.G and Bartolucci.F [2006]proposed an unbiased estimator for $E_L(Var_p(Y_s))$ defined as

$$Var_L(\overline{Y}_s) \quad = \tfrac{1}{kn^2}\hat{\beta}_i^T X^T N \hat{\beta}_i - t\hat{\delta}_{Li}^2 + \tfrac{1}{kn^2} tr(N\Omega)\hat{\delta}_{Li}^2 \tag{3.15}$$

where $\hat{\beta}_i$ is the OLS estimates for β for the i^{th}sample and $\hat{\sigma}^2_{Li}$ is the model unbiased estimate for $\hat{\sigma}^2_L$ which is define as

$$\hat{\sigma}^2_{Li} = \frac{(Y_i - X_i\hat{\beta}_i)^T\Omega_i^{-1}(Y_i - X_i\hat{\beta}_i)}{n - Rank(X)}$$

where Ω_i is a sub matrix of Ω corresponding to the ith sample In equation 3.15 $t\hat{\sigma}^2_{Li}$ is the bias correction term with

$$t = \frac{1}{(kn)^2}\sum_{i=1}^{k} tr(P_i^T X^T N X P_i \Omega_i)$$

and

$$P_i = (X_i^T\Omega_i^{-1}X_i)^{-1}X_i^T\Omega_i^{-1}$$

we assume that $(X_i^T\Omega_i^{-1}X_i)^{-1}$ exists X.Li [2006] shows that under an asymptotic framework in which $N \longrightarrow \infty$ and a set of assumptions

$$var(Y_s) - E_L(Var_p(\overline{Y}_s)) \quad = O_p(\tfrac{1}{\sqrt{N}}) \tag{3.16}$$

$$\widehat{V}_L(\overline{Y}_s) - E_L(Var_p(\overline{Y}_s)) \quad = O_p(\tfrac{1}{\sqrt{n}}) \tag{3.17}$$

$$\widehat{V}_L(\overline{Y}_s) - Var_p(\overline{Y}_s) \quad = O_p(\tfrac{1}{\sqrt{n}}) \tag{3.18}$$

equation 3.17 suggests that $\widehat{V}_L(\overline{Y}_s)$ is a consistent estimator for $E_L(Var_p(\overline{Y}_s))$ and equation 3.16 also suggests that the conditional variance $Var_p(\overline{Y}_s)$ converges to $E_L(Var_p(\overline{Y}_s))$ in probability. Therefore $Var_L(\overline{Y}_s)$ is the unbiased estimator for $Var_p(\overline{Y}_s)$ For more literature and details on this result (see X.Li [2006])

3.6 Assumptions used in developing the estimator in the current study

To prove the convergence properties of the proposed estimator,the study adopt a theoretical framework in which both the population size N, the sample size n and the sampling interval tend to infinity.A sample is the selected as described in section 3.1.

We make the following additional assumption on the study variable,the design and the smoothing methodology:

A1: The errors $\varepsilon_j' s$ are independent with a mean zero and variance $w_j \delta_{NP}^2$ and compact support, uniformly for all N

A2: For each N, we consider the $x_j' s$ as fixed with respect to the super-population model $N.P$. The $x's$ are independent and identically distributed. $F(x) = \int_{-\infty}^{x} f(t)dt$, where $f(.)$ is the density function with compact support $[a_x, b_x]$ and $f(x) > 0$ for all $X \epsilon [a_x, b_x]$

A3: The sample size n and sampling interval k are positive integers with $nk = N$ we assume that $n, N \to \infty$ and allow $k = O(1)$ or $k \to \infty$

A4: As $n \to \infty$, we assume $h^* \to 0$ and $nh^* \to \infty$ where $h^*$is h_m

A5: The kernel function $k(.)$ is a compactly supported, bounded, symmetric kernel with $\int u^{q+1} k(u)du = u_{q+1}(k)$ assume that $u_{p+1}(k) \neq 0$

A6: The $(p+1)^{th}$derivative of the mean function $\mu(.)$ exists and is bounded on $[a_x, b_x]$

3.7 Proposed Estimator for variance of Systematic Sample Mean

In this study a consistent variance estimator of the systematic sample means is proposed under a nonparametric model using local polynomial regression as the method of estimation. In the estimation procedure, let $X = (X_1, \ldots, X_N)$ be the univariate auxiliary variable, then,the nonparametric superpopulation model is given by equation 3.7 where

$$E_p(Y_j|X_j = x_j) = \mu(x_j)$$

and

$$Var_{NP}(\varepsilon) = \delta^2_{NP}$$

$$diag(w_1, w_2, \ldots, w_N) = \delta^2_{NP}\Omega$$

Let $\mu(.)$ be the a continuous and bounded function and define
$\mu = (\mu(x_1), \mu(x_2), \ldots, \mu(x_N))$ we assume that $w_j's$ are bounded and positive where

$$j = 1, 2, 3, \ldots, N$$

Under model 3.7, the expected value of $var_p(\overline{Y}_s)$ is

$$E_{NP}(var_p(\overline{Y}_s)) = \frac{1}{kn^2}\mu^T N\mu + \frac{1}{kn^2}tr(N\Omega)\delta^2_{NP} \tag{3.19}$$

To estimate $E_{NP}(var_p(\overline{Y}_s))$, the following local polynomial regression estimator for variance of systematic sample means is proposed:

$$\widehat{V}_{NP}(\overline{Y}_s) \quad = \tfrac{1}{kn^2}(\hat{\mu}_i^T N\hat{\mu}_i) + \tfrac{1}{kn^2}tr(N\Omega)\hat{\delta}^2_{NP} \tag{3.20}$$

here

$$\hat{\delta}^2_{NP} = \frac{(Y_i - \hat{\mu}_i)^T \Omega_i^{-1} (Y_i - \hat{\mu}_i)}{n}$$

and

$$\hat{\mu}_i = (\hat{\mu}(x_1), \ldots, \hat{\mu}(x_n))$$

where $\hat{\mu}(x_j)$ is the local polynomial regression estimator obtained from the jth sample

$$\hat{\mu}_i = e_1^T (X_{ij}^T W_{ij} X_{ij})^{-1} X_{ij}^T W_{ij} Y_i$$

Where e is the $(p \times 1) \times 1$ vector of the identity matrix having 1 in the first entry and all other entries 0 and p denotes the degree of local polynomial regression.

$$X_{ij} = \begin{pmatrix} 1 & (x_1 - x_j) & . & . & (x_1 - x_j)^p \\ . & . & . & . & . \\ . & . & . & . & . \\ . & . & . & . & . \\ 1 & (x_n - x_j) & . & . & (x_n - x_j)^p \end{pmatrix}$$

$$W_{ij} = diag(K(\frac{x_i - x_j}{h})), i = 1, 2, \ldots, n$$

where h is the bandwidth and $K(.)$ the kernel function

In developing the current estimator, reference is made to Wand and Jones [1995] version of the local polynomial regression estimator.

The estimator proposed by Montanari.G and Bartolucci.F [1998],Montanari.G and Bartolucci.F [2006] included the bias correction term which was asymptotically negligible and that estimator developed by X.Li and Opsomer [2010] considered the variance of the error term of the nonparametric model being heteroscedastic and which was estistimated locally by local polynomial technique,this study considers a more applicable model in which the mean function is unknown and is estimated from the sample and the condition on the variance function used by X.Li and Opsomer

[2010] is relaxed and also the bias correction term is not included in developing the current estimator since its asymptotically negligible.

under assumption $A_1 - A_6$, the design variance is model consistent for the anticipated variance in the sense that

$$var_p(\overline{Y}_s) \quad = E_p(var_p(\overline{Y}_s)) = O\left(\frac{1}{\sqrt{N}}\right) \tag{3.21}$$

and the NP variance estimator is model consistent for the anticipated variance for the design variance in the sense that

$$\widehat{V}_{NP} - E(var_p(\overline{Y}_s)) \quad = O_p(h_m^{p+1}) + O_p\left(\frac{1}{\sqrt{nh_m}}\right) \tag{3.22}$$

$$\widehat{V}_{NP} - var_p(\overline{Y}_s) \quad = O_p\left(h_m^{p+1}\right) + O_p\left(\frac{1}{\sqrt{nh_m}}\right) \tag{3.23}$$

the best band width should should satisfy the condition

$$h_m^{p+1} = O\left(\frac{1}{\sqrt{nh_m}}\right)$$

which leads to

$$h_m = C_n^{-\frac{1}{2(p+3)}}$$

the usual optimal rate for local polynomial regression . The bandwidth selection procedures such as cross validation or plug-in method can be used in this case. This study provides an outline for the proof of equation 3.22. The proof for equation 3.21, and 3.23 see X.Li [2006]

3.7.1 Proof of Equation 3.22

From equation 3.22,

$$\begin{aligned}\widehat{V}_{NP} - E(var_p(\overline{Y}_s)) &= \frac{1}{kn^2}(\hat{\mu}_i^T N\hat{\mu}_i - \mu^T N\mu) + \frac{1}{kn^2}tr(N\Sigma)(\hat{\sigma}^2_{NPi} - \sigma^2_{NP}) \\ &= * + **\end{aligned} \tag{3.24}$$

The first term on the right hand side of equation 3.24 can be written as

$$\begin{aligned}\frac{1}{kn^2}(\hat{\mu}_i^T N\hat{\mu}_i - \mu^T N\mu) &= \frac{1}{kn^2}(\hat{\mu}_i - \mu)^T N(\hat{\mu}_i - \mu) + \frac{1}{kn^2}\mu^T N(\hat{\mu}_i - \mu) + \frac{1}{kn^2}(\hat{\mu}_i - \mu)^T \mu N \\ &= \frac{1}{kn^2}(\hat{\mu}_i - \mu)^T N(\hat{\mu}_i - \mu) + \frac{2}{kn^2}\mu^T N(\hat{\mu}_i - \mu) \\ &= A + 2B\end{aligned}$$

note that

$$\hat{\mu}^T N(\hat{\mu}_i - \mu) = \hat{\mu}^T N\mu$$

because they are both scalars. By definition of matrix N, we can write A as

$$\begin{aligned}A &= \frac{1}{k}\sum_{i=1}^{k}[\frac{1}{n}\sum_{j\in s_i}(\hat{\mu}(x_i) - \mu(x_i)) - \frac{1}{N}\sum_{j\in s_i}(\hat{\mu}(x_i) - \mu(x_i))]^2 \\ &= \frac{1}{k}\sum_{i=1}^{k}\{[\frac{1}{n^2}\sum_{j\in s_i}(\hat{\mu}(x_j) - \mu(x_j))]^2 + \frac{1}{N^2}[\sum_{j\in U}(\hat{\mu}(x_j) - \mu(x_j))]^2 - \frac{2}{nN}\sum_{j\in s_i}(\hat{\mu}(x_j) - \mu(x_j))\sum_{j\in U}(\hat{\mu}(x_j) \\ &= \frac{1}{k}\sum_{i=1}^{k}(a + b + c)\end{aligned}$$

where

$$a = \frac{1}{n^2}\sum_{j\in s_i}(\hat{\mu}(x_j) - \mu(x_j))]^2$$

$$b = \frac{1}{N^2}[\sum_{j\in U}(\hat{\mu}(x_j) - \mu(x_j))]^2$$

$$c = -\frac{2}{nN}\sum_{j\in s_i}(\hat{\mu}(x_j) - \mu(x_j))\sum_{j\in U}(\hat{\mu}(x_j) - \mu(x_j))$$

note that

$$\hat{\mu}(x_j) - \mu(x_j) = S_{ij}Y_j - \mu(x_j)$$

$$= S_{ij}(\mu_i - \varepsilon_j) - \mu(x_j)$$

$$= b_i(x_j) + S_{ij}\varepsilon_j$$

Here S_{ij} is the smoother matrix and $S_{ij} = e_1^T(X_{ij}^T W_{ij} X_{ij})^{-1} X_{ij}^T W_{ij}$ where X_{ij} and W_{ij} are defined. In this case for simplicity we will use $K_{ij}$to denote $K(\frac{x_i - x_j}{h})$ now expand the parentheses in (a) we have

$$E(a) = \frac{1}{n^2}\sum_{j\epsilon s_i} b_i^2(x_j) + \frac{1}{n^2}E\left(\sum_{j\epsilon s_i} S_{ij}\varepsilon_i\varepsilon_i^T S_{ij}^T\right) + \frac{1}{n^2}E\left(\sum_{j\epsilon s_i l\epsilon s_i, j\neq l} b_i(x_j)b_i(x_l)\right)$$
$$+ \frac{1}{n^2}E\left(\sum_{j\epsilon s_i l\epsilon s_i, j\neq l} S_{ij}\varepsilon_i\varepsilon_i^T S_{ij}^T\right) \tag{3.25}$$

The right hand side of equation 3.25 contains four terms. the researcher will calculate them one by one. (i) first let us investigate $\frac{1}{n^2}\sum b_i^2(x_j)$. The technique similar to Wand and Jones [1995] is used. let

$$\mu_i = (\mu(x_1), \ldots, \mu(x_n))$$

then by Taylor theorem

$$\mu_i = X_{ij}\begin{pmatrix}\mu(x_j)\\ D_{\mu(x_j)}\end{pmatrix} + R_{\mu(x_j)}$$

where

$$D_{\mu(x_j)} = (\mu^{'}(x_j), \frac{1}{2}\mu^{"}(x_j), \ldots, \frac{1}{p}\mu^p(x_j))$$

and $R_{\mu(x_j)}$ is a vector of Taylor series remainder terms so

$$\begin{aligned} b_i(x_j) &= S_{ij}\mu_i - \mu(x_j) \\ &= S_{ij}R_{\mu(x_j)} \\ &\equiv e_1^T Z_j^{*-1} T \end{aligned}$$

under assumptions A_2 and A_3 by lemma 2(ii) Bredit and Opsomer(2000) for a certain point x_j there are atleast $q+1$ points in the interval $[x_j - h, x_j + h]$. so Z_j^* is invertible.
Lemma 1: Assume that the kernel function $K_{ij}$is bounded above, then

$$\frac{1}{nh}\sum_{i=1}^{n} K_{ij}\left(x_i - x_j\right)^r = O\left(h^r\right)$$

where

$$r = 0, 1, 2, \ldots$$

the proof of lemma 1 is provided by X.Li [2006]. Thus suppose assumption A_4 holds, by lemma 1, we have

$$z_{stj} = O\left(h^{s+t-2}\right)$$

and

$$t_{kj} = O\left(h^{k+p}\right)$$

$$b_i(x_j) = e_1^T \begin{pmatrix} O(1)...O(h^p) \\ O(h^p)...O(h^{2p}) \end{pmatrix}^{-1} . \begin{pmatrix} O(h^{p+1}) \\ O(h^{2p+1}) \end{pmatrix}$$

note that the order of a matrix is the same as its inverse so

$$b_i(x_j) = O\left(h^{p+1}\right) + O\left(h^{3p+1}\right) = O\left(h^{p+1}\right) \tag{3.26}$$

thus

$$\frac{1}{n^2}\sum_{j\epsilon s_i} b^2(x_j) = \frac{1}{n^2}\sum_{j\epsilon s_i} O\left(h^{2p+2}\right) = O\left(\frac{h^{2p+2}}{n}\right) \tag{3.27}$$

(ii) secondly, now compute

$$\frac{1}{n^2}E\left(\sum_{j\epsilon s_i} S_{ij}\varepsilon_i\varepsilon_i^T S_{ij}^T\right)$$

in 3.25

$$\begin{aligned}\frac{1}{n^2}E\left(\sum_{j\epsilon s_i} S_{ij}\varepsilon_i\varepsilon_i^T S_{ij}^T\right) &= \frac{1}{n^2}E\left(\sum_{j\epsilon s_i} S_{ij}\Sigma_{ij}S_{ij}^T\right)\\ &= e_1^T(X_{ij}^TW_{ij}X_{ij})^{-1}X_{ij}^TW_{ij}\Sigma_{ij}X_{ij}(X_{ij}^TW_{ij}X_{ij})^{-1}e_1\\ &= \frac{1}{n^2}\sum_{j\epsilon s_i} e_1^TZ_j^{*-1}C_j^*Z_j^{*-1}e_1\end{aligned}$$

where Σ_i is the variance covarance matrix of the model 3.7 and

$$\Sigma_i = diag(w_1, w_2, \ldots, w_n)$$

$$C_j^* = \begin{pmatrix} c_{11j}\ldots c_{1(p+1)j} \\ c_{(p+1)1j}\ldots c_{(p+1)(p+1)j}\end{pmatrix}$$

with

$$c_{slj} = \frac{1}{n^2h^2}\sum_{i=1}^{n} K_{ij}^2 w_i(x_i - x_j)^{s+t-2} = O\left(\frac{h^{s+t-3}}{n}\right)$$

by lemma 1 X.Li [2006] shows that

$$e_1^TZ_j^{*-1}C_j^*Z_j^{*-1}e_1 = O\left(\frac{1}{nh}\right)$$

and thus

$$\frac{1}{n^2}E(\sum_{j\epsilon s_i} S_{ij}\varepsilon_i\varepsilon_i^T S_{ij}^T) = O\left(\frac{1}{n^2h}\right) \tag{3.28}$$

(iii) Thirdly we now calculate

$$\frac{1}{n^2}E(\sum_{j\epsilon s_i}\sum_{l\epsilon s_i, j\neq l} b_i(x_j)b_i(x_l))$$

in 3.25 using the results in 3.26 we get

$$\frac{1}{n^2}E(\sum_{j\epsilon s_i}\sum_{l\epsilon s_i, j\neq l} b_i(x_j)b_i(x_l)) = O\left(h^{2p+2}\right) \tag{3.29}$$

(iv) the last term on the right hand side of 3.25 is

$$\frac{1}{n^2}E(\sum_{j\epsilon s_i}\sum_{l\epsilon s_i, j\neq l} S_{ij}\varepsilon_i\varepsilon_i^T S_{ij}^T))$$

and X.Li [2006] shows that

$$\frac{1}{n^2}E(\sum_{j\epsilon s_i}\sum_{l\epsilon s_i, j\neq l} S_{ij}\varepsilon_i\varepsilon_i^T S_{ij}^T)) = O\left(\frac{1}{n}\right) \tag{3.30}$$

Assumption $A3$implies that $nh \to \infty$, and by 3.27,3.28 and 3.30

$$E(a) = O\left(h^{2p+2}\right) + O\left(\frac{1}{n}\right) \tag{3.31}$$

simillaly we can calculate $E(b)$ and $E(c)$. under $A3$ $nh \to \infty$ so

$$E(b) = O\left(h^{2p+2}\right) + O\left(\frac{1}{N}\right) \tag{3.32}$$

and

$$E(c) = O\left(h^{2p+2}\right) + O\left(\frac{1}{n}\right) \tag{3.33}$$

Also not that $A > 0$ and $|A| = A$ thus by 3.31, 3.32 and 3.33

$$E|A| = E(A) = \frac{1}{k}\sum_{i=1}^{k}(E(a) + E(b) + E(c)) = O\left(h^{2p+2}\right) + O\left(\frac{1}{n}\right)$$

which implies that

$$A = O_p\left(h^{2p+2}\right) + O_p\left(\frac{1}{n}\right)$$

Next using a similar approach to that of A

$$B = O_p\left(h^{p+1}\right) + O_p\left(\frac{1}{\sqrt{n}}\right)$$

Thus

$$\begin{aligned} * &= A + 2B \\ &= O\left(h^{p+1}\right) + O_p\left(\frac{1}{\sqrt{n}}\right) \end{aligned} \tag{3.34}$$

now let us calculate $**$ in 3.24

$$** = \frac{1}{kn^2} tr(D\Sigma)((\hat{\sigma}^2_{NPi} - \sigma^2_{NP}))$$

in 3.24 and

$$\hat{\sigma}^2_{NPi} - \sigma^2_{NP} = \frac{1}{n}\sum_{j=1}^{n} \frac{(Y_j - \mu(x_j))^2}{w_j} - \sigma^2_{NP} + \frac{1}{n}\sum_{j=1}^{n} \frac{(\hat{\mu}(x_j) - \mu(x_j))^2}{w_j} + \frac{2}{n}\sum_{j=1}^{n} \frac{(Y_j - \hat{\mu}(x_j))(\mu(x_j) - \mu(x_j))}{w_j}$$

X.Li [2006] shows that

$$\frac{2}{n}\sum_{j=1}^{n} \frac{(Y_j - \mu(x_j))^2}{w_j} - \sigma^2_{NP} = O_p\left(\frac{1}{\sqrt{n}}\right) \tag{3.35}$$

$$\frac{1}{n}\sum_{j=1}^{n} \frac{(\hat{\mu}(x_j) - \mu(x_j))^2}{w_j} = O_p\left(h^{2p+2}\right) + O_p\left(\frac{1}{nh}\right) \tag{3.36}$$

and

$$\frac{2}{n}\sum_{j=1}^{n} \frac{(Y_j - \hat{\mu}(x_j))(\mu(x_j) - \mu(x_j))}{w_j} = O_p\left(h^{p+1}\right) + O_p\left(\frac{1}{\sqrt{nh}}\right) \tag{3.37}$$

since

$$\frac{1}{kn^2}tr(D\Sigma) = O\left(1\right)$$

and by 3.35,(3.36) and 3.37, the following equation is obtained

$$\frac{1}{kn^2}tr(D\Sigma)((\hat{\sigma}^2_{NPi} - \sigma^2_{NP})) = O_p\left(h^{p+1}\right) + O_p\left(\frac{1}{\sqrt{nh}}\right) \tag{3.38}$$

Therefore by (3.34) and (3.36)

$$\widehat{V}_{NP} - E(var_p(\overline{Y}_s)) = O_p\left(h^{p+1}\right) + O_p\left(\frac{1}{\sqrt{nh}}\right)$$

hence the results.

3.8 Simulation study

To further investigate the statistical properties of the above variance estimators,a simulation study is was performed. For simplicity,the researcher considered the case where there was only one auxiliary variable x. It is also assumed that the errors are independently and normally distributed with homogeneous variances. Two super-population models are examined. One is the linear model

$$y_j \quad = 2x_j + \varepsilon_j \tag{3.39}$$

where

$$j = 1, 2, \ldots, N$$

and

$$\varepsilon_j \sim N(0, \delta_1^2)$$

The quadratic model

$$y_j \quad = 1 + 2(x_j - 0.5)^2 + \varepsilon_j \tag{3.40}$$

$$\varepsilon_j \sim N(0, \delta_2^2)$$

The bigger the R^2, the bigger the predictive power of the model. The two levels of R^2that are achieved are $R^2 = 0.75$"$precise$" $model$ and $R^2 = 0.25$ the $diffuse\,model$.

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i\epsilon U} \varepsilon_j^2}{\sum_{i\epsilon U} (Y_i - Y_u)^2}$$

To draw a systematic sample, the population first need to be sorted. Three ways are considered: (1) Sort by auxiliary variable x; (2) Sort by z_1 , where $z_{1j} = x_j + \eta_{1j}$ and $\eta_{1j \sim N(0,\delta^2_{\eta 1})}$. Choose $\delta^2_{\eta_1}$ to make $R^2_{z1} = 0.75$ (3) Sort by z_2 , where $z_{2j} = x_j + \eta_{2j}$ and$\eta_{2j} \sim N(0, \delta^2_{\eta_2})$. Choose $\delta^2_{\eta_2}$ to make $R^2_{z2} = 0.25$. Populations of size $N = 2000$ is generated. To achieve this, 2000values of model variable x from the uniform distribution on $[0, 1]$ and 2000 values of error ε from $N(0, 1)$ were generated. Then 2000 values of response variable y computed by model 3.39 and 3.40. Two systematic samples of size $n = 500$ and $n = 100$, with corresponding sampling intervals $k = 4$ and $k = 20$ are considered respectively. To draw a systematic sample, the data first sorted, either by x or z, from the smallest to the largest, then randomly choose an observation from the first k observations, say the ith one. Then, the selected sample consists of the observations with the following subscripts: $i, i + k, \ldots, i + (n - 1)k$. For each simulation, the corresponding $Var_p(\overline{Y}_s)$,
$E_p[Var_p(\overline{Y}_s)], \widehat{V}_{NP}(\overline{Y}_s), \widehat{V}_{NP}(\overline{Y}_s), \widehat{V}_{OL}(\overline{Y}_s)$ and$V_{NO}(\overline{Y}_s)$ is calculated. For , $\widehat{V}_{NP}(\overline{Y}_s)$ it is calculated using two bandwidth values : $h = 0.50$ and $h = 0.25$. each simulation setting is repeated $B = 10,000$ times. The researcher then compare the performance of the nonparametric variance estimator $\widehat{V}_{NP}(\overline{Y}_s)$ with the overlapping differences

estimator $\widehat{V}_{OL}(\overline{Y}_s)$, the non-overlapping differences estimator $\widehat{V}_{NO}(\overline{Y}_s)$, which are recommended by Wolter [2007] and the simple random sampling estimator $\widehat{V}_{SRS}(\overline{Y}_s)$. The relative bias ($RB$) and the mean squared error (MSE) are calculated. Let $\widehat{V}$ represent $\widehat{V}_{SRS}(\overline{Y}_s)$, $\widehat{V}_{NP}(\overline{Y}_s)$, $\widehat{V}_{OL}(\overline{Y}_s)$, and $\widehat{V}_{NO}(\overline{Y}_s)$:

$$RB = \frac{E_{MP}(\widehat{V}) - E_M(Var_p(\overline{Y}_s))}{E_M(Var_p(\overline{Y}_s))}$$

$$MSE = E_{MP}(\widehat{V} - E_M(Var_p(\overline{Y}_s)))^2$$

$$MSPE = E_{MP}(\widehat{V} - vVar_p(\overline{Y}_s))^2$$

where E_M denotes the expectation under the superpopulation model M, and E_{MP} denotes the expectation under both the model and design. In this case $MSPE$ is not considered.

Chapter 4

SIMULATION AND DISCUSSION OF RESULTS

4.1 RESULTS AND DISCUSSION

Table 4.1 report the relative bias of $\widehat{V}_{NP}(\overline{Y}_s), \widehat{V}_{NO}(\overline{Y}_s), \widehat{V}_{OL}(\overline{Y}_s)$ and$V_{SRS}(\overline{Y}_s)$ for the sample of size $n = 500$ for different sorting variables with homoscedastic errors. The relative biases for non parametric estimators are computed at different bandwidth. The results from table show that given a proper bandwidth is chosen, non parametric estimator performs well overall than other three estimators resulting to smaller biases with most biases being less than zero. Especially when the super-population model is linear, $\widehat{V}_{NP}(\overline{Y}_s)$ tends to favor bigger bandwidth. This is because local linear regression was used in the calculation of $\widehat{V}_{NP}(\overline{Y}_s)$. The bigger bandwidth results in more points in the neighborhood of x_j and because the local polynomial regression is local linear which is correct one for this population with linear trend, so having more points will increase the precision of each local linear regression.

When the super-population model is quadratic, it tends to favor small bandwidth. This is because, as discussed above, for parametric estimation, linear regression will

Mean Function		LINEAR			QUADRATIC		
Sorting variable		1	0.75	0.25	1	0.75	0.25
	$h = 0.1$	-1.9370	-0.9182	-0.7825	-0.9939	-0.9739	-0.9196
V_{NP}	$h = 0.25$	-0.9558	-0.5263	-0.6926	-0.9778	-0.9397	-0.9046
	$h = 0.5$	-0.9459	-0.8409	-0.8703	-0.9519	-0.9287	-0.8562
	V_{OL}	-0.2984	0.1600	0.4732	0.1110	0.3430	-2.0250
	V_{NO}	-0.3035	0.1624	0.4654	0.1197	0.3413	-1.0196
	V_{SRS}	0.3009	0.1685	4.5830	12.0362	0.4762	-1.0264

Table 4.1: Relative Bias for $\widehat{V}_{NP}$(with bandwidth $h = 0.1, 0.25, 0.5$), $\widehat{V}_{NO}(\overline{Y}_s), \widehat{V}_{OL}(\overline{Y}_s)$ and $V_{SRS}(\overline{Y}_s)$ with $n = 500$

not estimate quadratic trend well. The wider the neighborhood, the more likely a quadratic trend will be seen there. Therefore local linear regression on that neighborhood could be bad. If the bandwidth is small, then the trend within each local interval will be approximated well by a linear trend.

The estimator based on simple random sampling $V_{SRS}(\overline{Y}_s)$ performed poorly, resulting in large biases in both cases as it overestimated the true variance. The bias of SRS can be both negative and positive unless the sorting variable is itself randomized over repeated systematic sampling draws.

It can also be seen that $\widehat{V}_{OL}(\overline{Y}_s)$and $\widehat{V}_{NO}(\overline{Y}_s)$ have smaller biases under linear and quadratic models when the population is sorted by x before drawing a systematic sample. This is because$\widehat{V}_{OL}(\overline{Y}_s)$and $\widehat{V}_{NO}(\overline{Y}_s)$ capture the population trend very well and thus very efficient. When the sorting variable is not related to the population that is sorting the population by z_1 and z_2, overlapping and non overlapping difference estimators cannot capture the population trend well hence resulting to large biases and MSE. Also $\widehat{V}_{OL}(\overline{Y}_s)$and $\widehat{V}_{NO}(\overline{Y}_s)$ do not depend on the auxiliary variable x, therefore the non-parametric estimator $\widehat{V}_{NP}(\overline{Y}_s)$ has the advantage of using auxiliary information to improve its precision under these circumstances. Table 4.2 reports the ratios of MSE for $\widehat{V}_{NP}(\overline{Y}_s)$ that is evaluated at two bandwidth values (h=0.25 and h=0.5) obtained by dividing the MSE of other estimators by the of $\widehat{V}_{NP}(\overline{Y}_s)$ evaluated at a bandwidth(h=0.1). The MSE measures the variability of

Mean Function		LINEAR			QUADRATIC		
Sorting variable		1	0.75	0.25	1	0.75	0.25
V_{NP}	$h = 0.25$	0.37	0.58	1.00	1.00	0.99	10.00
	$h = 0.5$	0.56	0.23	1.00	1.10	1.01	10.99
	V_{OL}	4.34	3.98	17.15	7.44	15.45	174.17
	V_{NO}	4.39	4.12	17.25	7.47	16.47	174.10
	V_{SRS}	43.86	38.69	27.02	17.32	17.33	172.40

Table 4.2: MSE for $\widehat{V}_{NP}$(with bandwidth $h = 0.25, 0.5$), $\widehat{V}_{NO}(\overline{Y}_s), \widehat{V}_{OL}(\overline{Y}_s)$ and $V_{SRS}(\overline{Y}_s)$ with $n = 500$ divided by MSE of $\widehat{V}_{NP}$ with bandwidth $h = 0.1$ and Homoscedastic errors

an estimator and smaller MSE values are normally desired.

From this study, it can be seen that, $\widehat{V}_{NP}(\overline{Y}_s)$ performs better than the $\widehat{V}_{OL}(\overline{Y}_s)$, $\widehat{V}_{NO}(\overline{Y}_s)$ and $V_{SRS}(\overline{Y}_s)$ as it has smaller MSE values in almost all the cases of linear and quadratic models.

Chapter 5

CONCLUSIONS AND RECOMMENDATIONS

The aim of this study was to develop design unbiased estimator of variance of the systematic means using local polynomial regression as the estimation technique. The study reveals that, the estimator based on non parametric model 3.7 using local polynomial regression regression as the estimation technique $\widehat{V}_{NP}(\overline{Y}_s)$ is a consistent estimator for the $V_p(\overline{y}_s)$.

Compared to other estimator discussed in Wolter [2007], the variance estimator $\widehat{V}_{NP}(\overline{Y}_s)$ under a non-parametric model 3.7 performed better in almost all the cases. Therefore, this estimator has proved to be consistent and unbiased in estimating the design variance of systematic sample mean.

Thus, in practice,the study recommends the use of non parametric estimator $\widehat{V}_{NP}(\overline{Y}_s)$ for estimating the variance of systematic sample mean over the estimators proposed by Wolter [2007].

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Appendix 1: Simulated data

$x \sim U(0,1)$

0.0074648280 0.4652723863 0.2040065359 0.4041413781 0.3134771322 [186] 0.1964791717 0.8517084182 0.1832381736 0.3370295111 0.0893442237 [191] 0.9724495129 0.7948795280 0.7550517912 0.4816733866 0.8940837129 [196] 0.6774629522 0.2748190823 0.6294147745

$e \sim N(0,1)$

-2.14404479 -0.22405058 -0.20712273 0.38667962 -0.72199715 -0.35664210 [7] -0.77800758 0.39359857 -0.98072575 -1.26418609 0.68477597 0.35950792 [13] 2.55301998 1.62209985 0.25262827 -1.35229141 0.41130518 0.33041775 [19] -0.78746258 0.61342582 1.08826611 -1.14089568 0.33530553 -0.06199713 [25] -0.19560126 -0.14371322 -1.73918411 -1.50391975 1.45982758 -0.52526320 [31] 0.70609463 0.44974827 2.17756906 -1.09058990 0.16371967 -1.56022313 [37] -0.44653825 0.56702257 0.39709369 1.26623848 -0.27653941 -0.34150155 [43] -0.76168156 1.43817427 -0.52321897 2.11037080 0.85406722 0.73133886 [49] -1.57978377 -0.61790168 0.20299106 0.59397685 1.29563301 1.36631415 [55] -0.28006702 -0.15524750 -0.23265628 -1.19330926 -0.60394838 -0.05038409 [61]

For the linear model

$y = 2x + e$

1.563538982 0.937551849 -0.026485042 1.293504966 0.751458837 [6] 0.443252221 0.317519537 1.154912527 0.940460022 -0.789040938 [11] 0.955424341 1.375429240 2.977087104 1.876937223 0.709172822 [16] 0.392650040 2.120061522 0.766053373 0.212501205 1.600813892 [21] 2.639265541 0.125474714 1.733034820 1.315174221 0.375693424 [26] 0.253405247 -1.125274812 0.006340719 3.042100123 1.211877087 [31] 1.125745420 1.057774010 2.299858772 0.549064585 1.335290390

for quadratic model

$y = 1 + 2(x - 0.5)^2$

-1.05605710 0.78900710 1.12855458 1.39102038 0.39008314 0.66337904 [7] 0.22655513 1.42208408 0.44356586 -0.12644978 1.95075287 1.35963466 [13] 3.71886931 2.89973351 1.40030018 -0.07482253 1.66247296 1.48967133 [19] 0.21253742 1.61350535 2.24006630